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AWS CLOUD PRACTITIONER

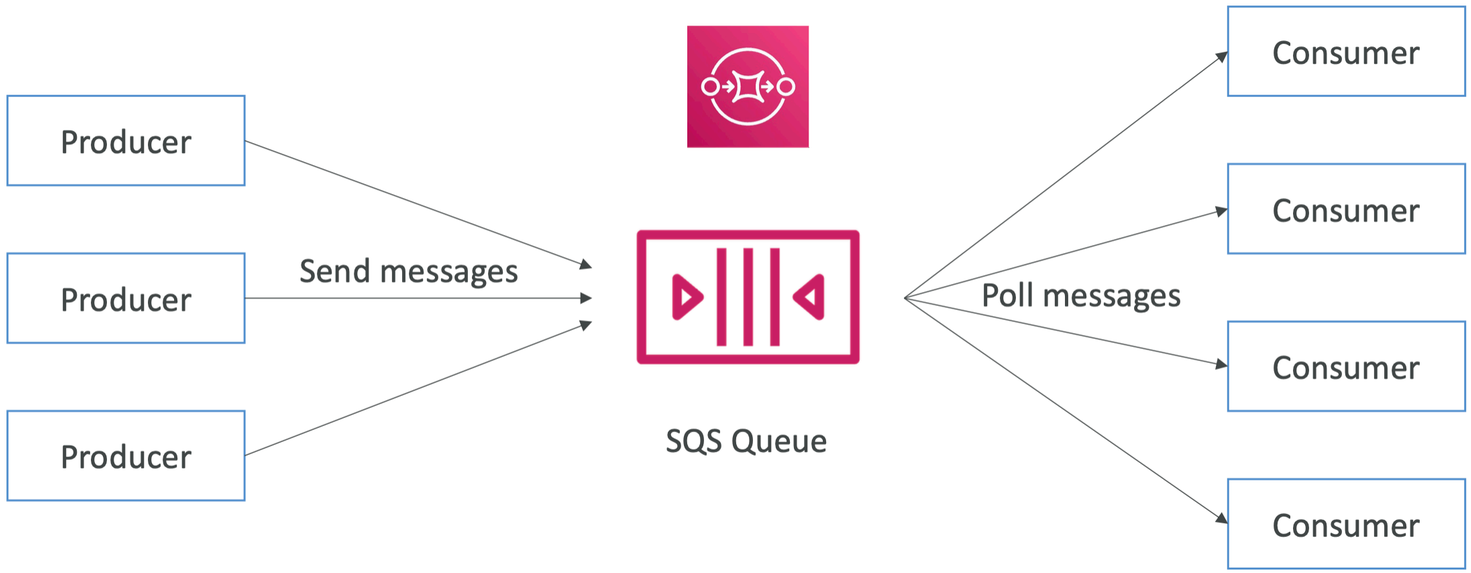
Amazon SQS, SNS and SES

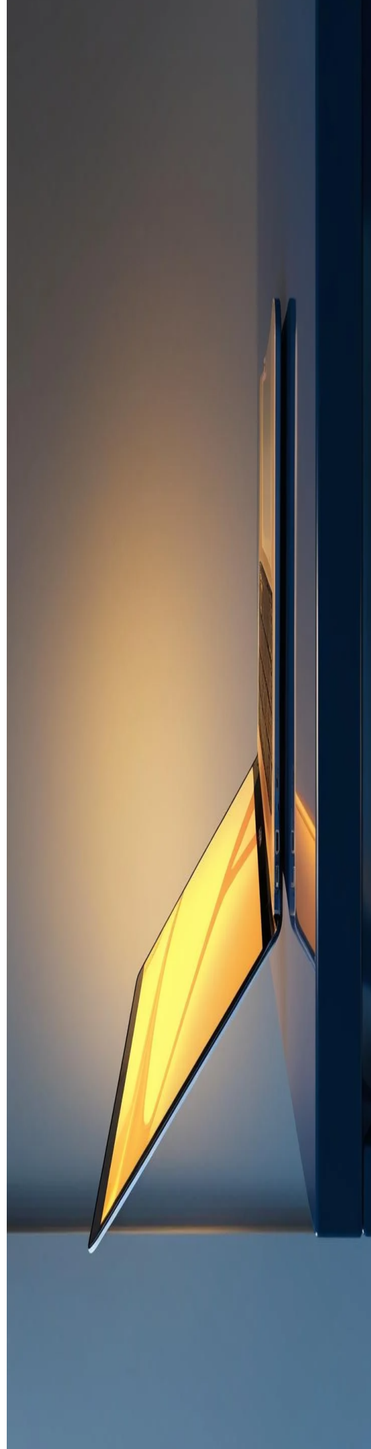
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27 October 2026

What is Amazon SQS?

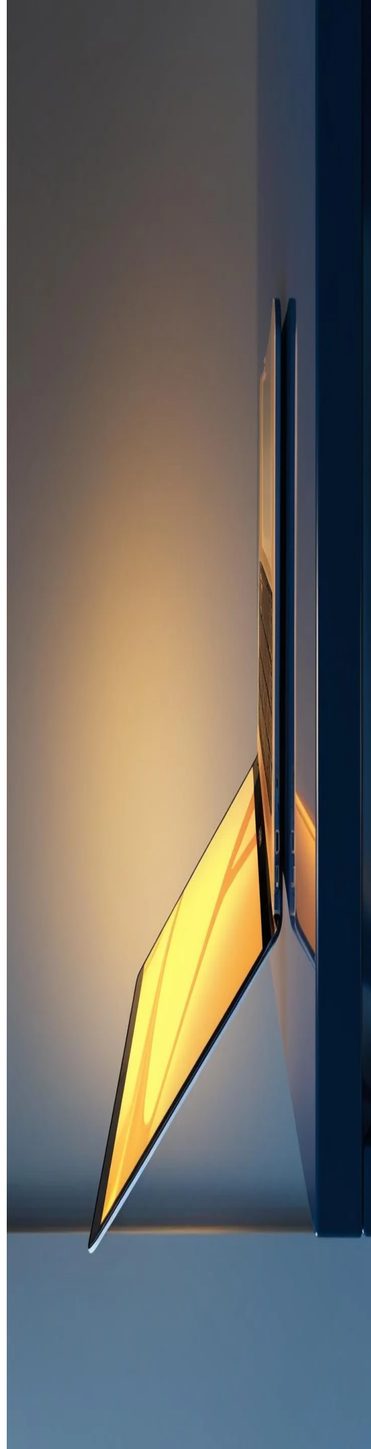
- Amazon Simple Queue Service (SQS) is a message queue service used by distributed applications to exchange messages through a polling model, and can be used to decouple sending and receiving components.
- Main idea: Store and process later.





Amazon SQS features

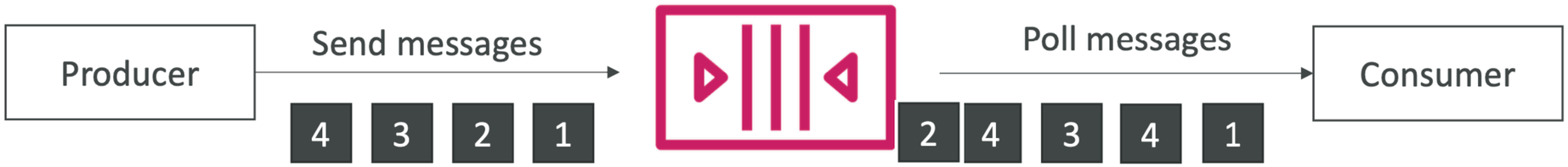
- **Fully managed message queue**: AWS manages the underlying infrastructure, reducing administrative overhead and configuration.
- **Serverless**: No servers need to be provisioned or maintained.
- **Asynchronous processing**: Producers can send messages without waiting for consumers to process them, helping decouple application components.
- **High scalability**: SQS can handle very large message volumes and scale with traffic without manual capacity configuration.
- **High message durability**: SQS stores messages redundantly across multiple Availability Zones within a Region to protect against infrastructure failures.
- **Simple APIs**: SQS provides web service APIs and AWS SDK support for many programming languages.
- **Two queue types**: Standard queues and FIFO (First-In, First-Out) queues support different ordering and delivery requirements.

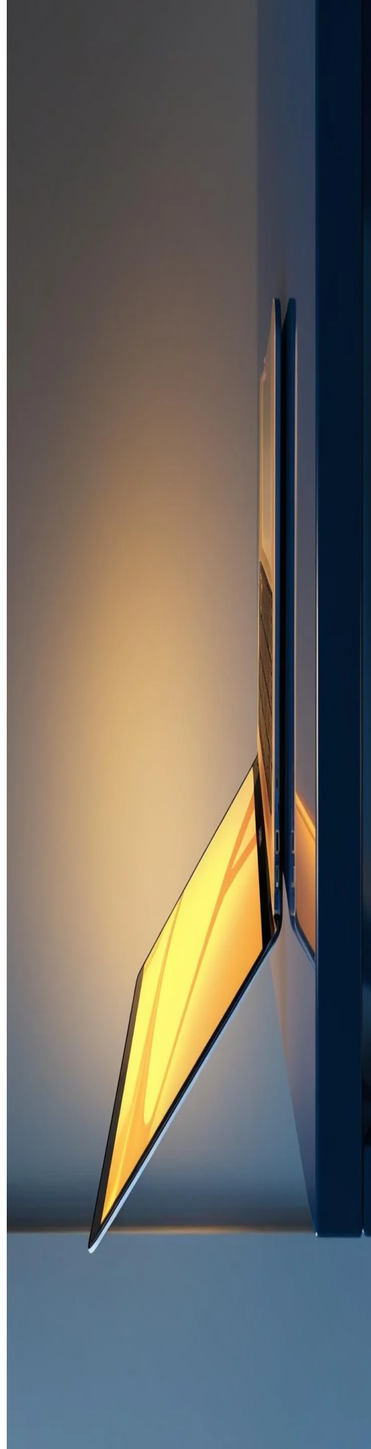


AWS SQS: Standard queue

Standard queue

- **Nearly Unlimited Throughput**: Supports a nearly unlimited number of transactions per second (TPS) per API action.
- **At-Least-Once Delivery**: Messages are delivered at least once, but duplicates may occasionally occur.
- **Best-Effort Ordering**: Messages are generally delivered in the order they are sent, but their order is not guaranteed.
- **Best for**: High-volume background tasks, parallel processing, and applications that prioritize scalability over strict message ordering.



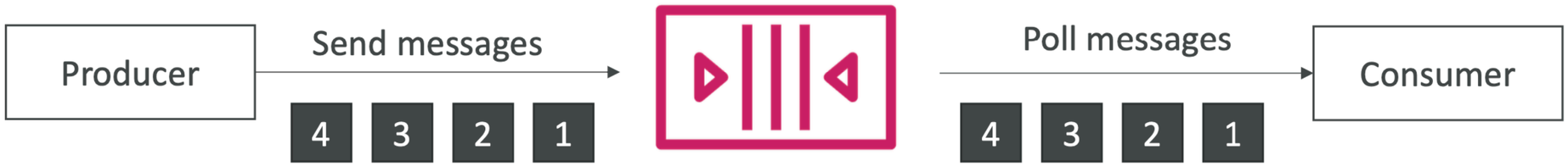


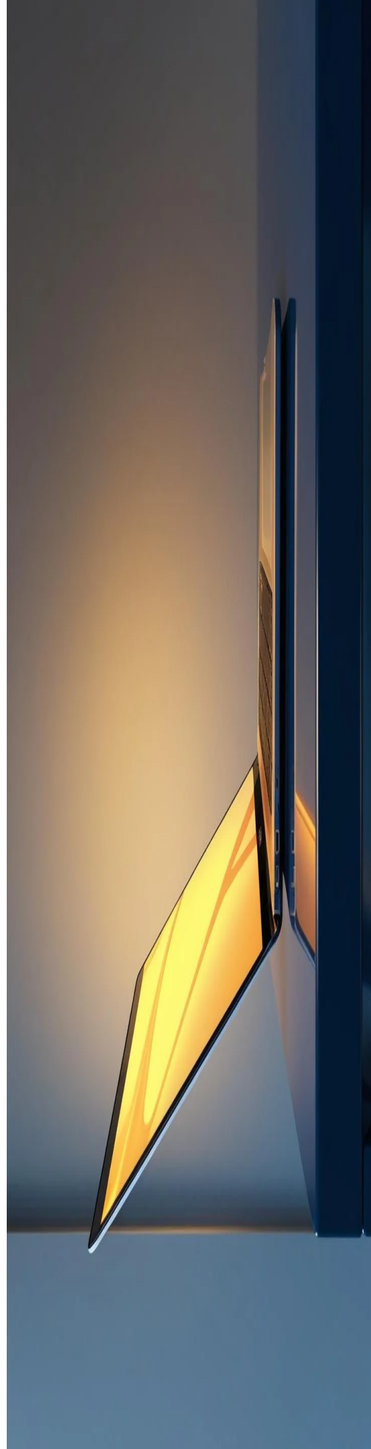
AWS SQS: FIFO queue

FIFO (First In First Out) queue

- **First-In-First-Out Delivery**: The order in which messages are sent and received is strictly preserved.
- **Exactly-Once Processing**: Deduplication prevents duplicate messages from being introduced within the deduplication interval. Consumers must still handle retries and processing failures appropriately.
- **Dead-letter queue (DLQ)**: Can be configured to isolate messages that repeatedly fail processing.
- **Throughput**: By default, FIFO queue support up to 300 messages per second without batching or 3,000 messages per second with batching in supported Regions and configurations. High-throughput mode can increase these limits.
- **Best for**: Order-sensitive workflows, such as financial transactions, inventory updates, and sequential order processing.

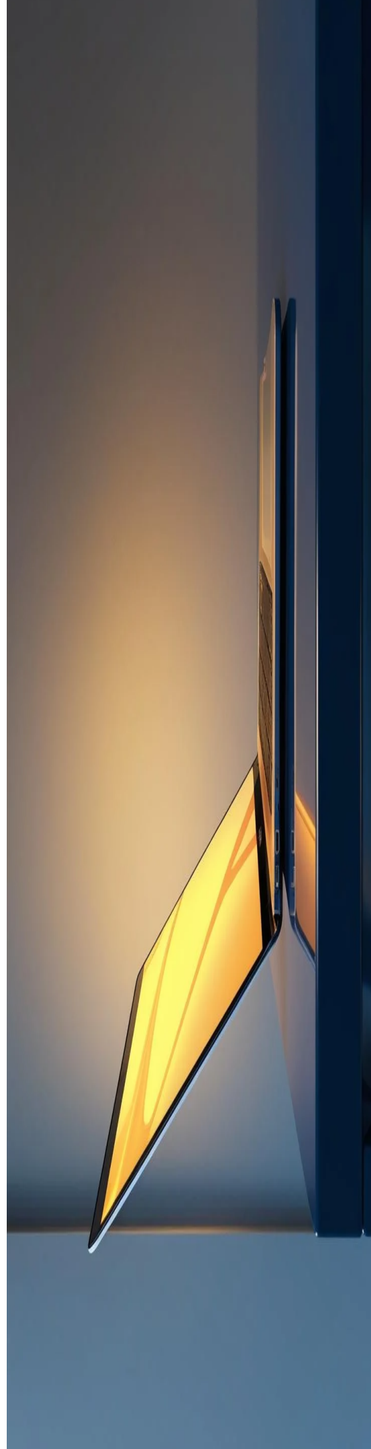
Remember: FIFO = ordered messages + deduplication.





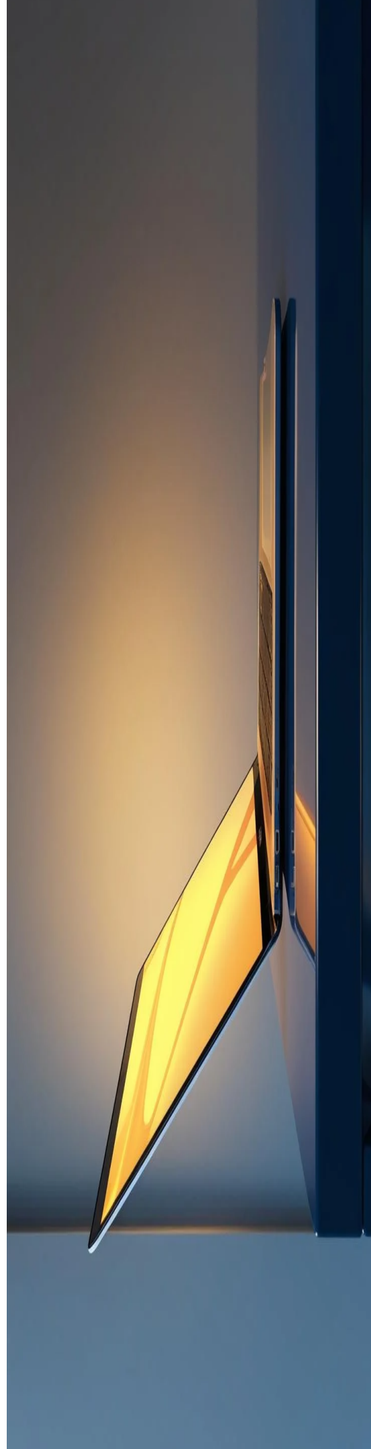
Amazon SQS: Uses Cases

- **Order processing systems**: Queue orders for background processing, such as payment verification, inventory updates, and shipping.
- **User sign-up workflows**: Decouple user registration from email verification so users can sign up while verification emails are processed asynchronously.
- **Serverless applications**: Trigger AWS Lambda functions or other serverless functions in response to queued events.
- **IoT data pipelines**: Buffer large volumes of sensor data and process it reliably without overwhelming downstream services.
- **Batch jobs**: Accumulate tasks in a queue and allow worker processes to handle them at their own pace.

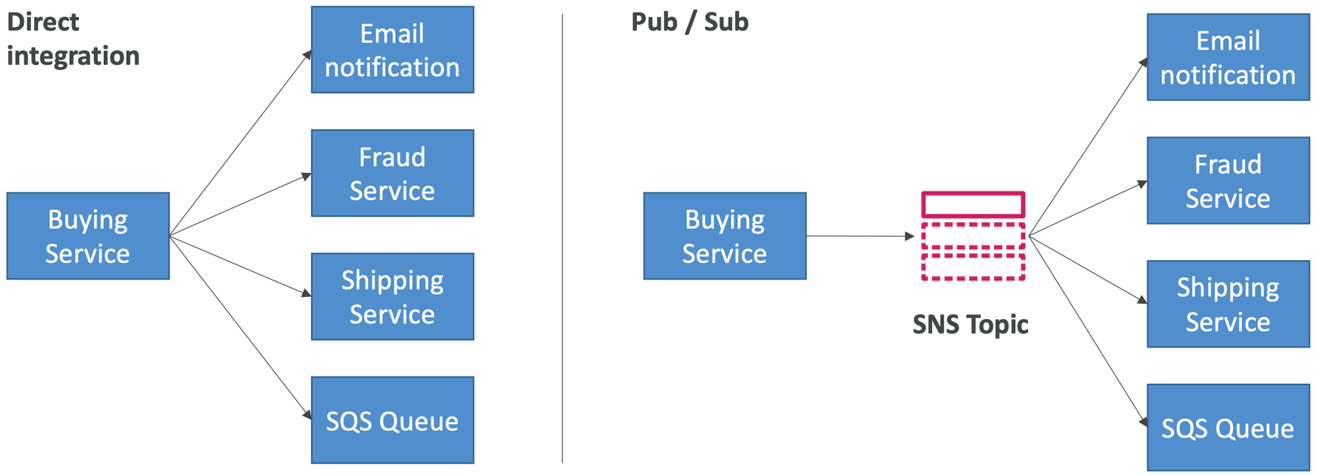


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What is Amazon SNS?

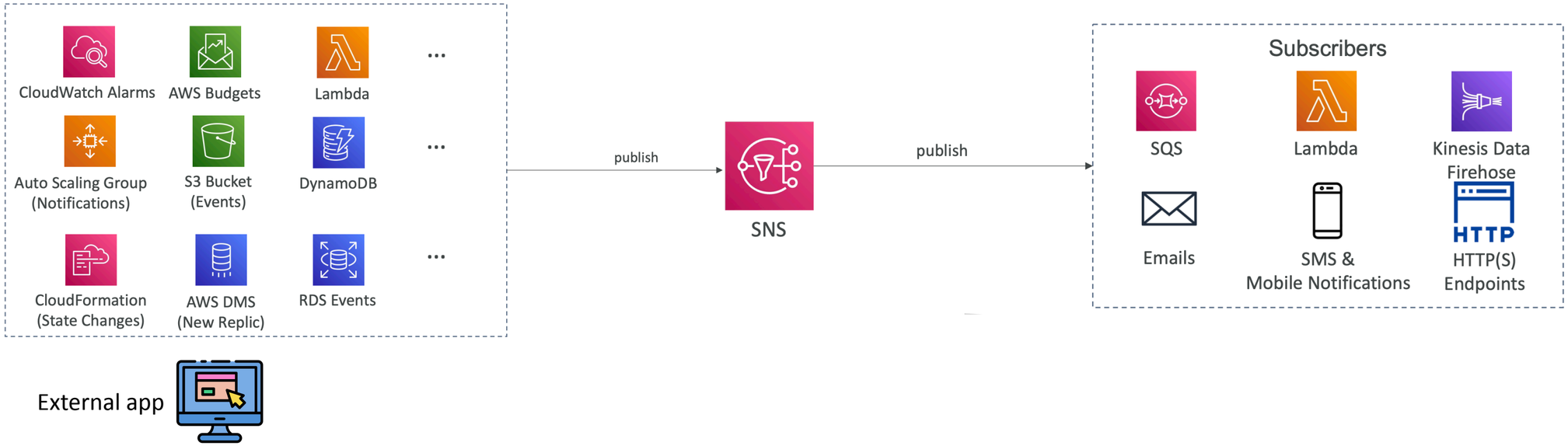


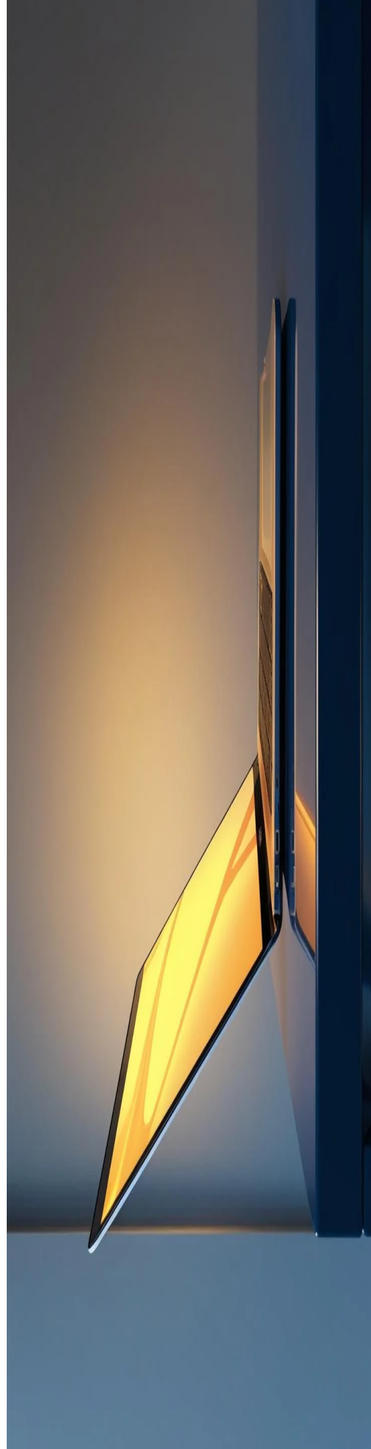
- Amazon Simple Notification Service (SNS) is a fully managed AWS messaging service that enables applications to distribute notifications to multiple subscribers using the Publish/Subscribe (Pub/Sub) model.
  - “push” mechanism: Amazon SNS automatically delivers messages to multiple subscribers, eliminating the need for them to periodically check or “poll” for updates.
- Main idea: Publish once, notify many.



Amazon SNS features

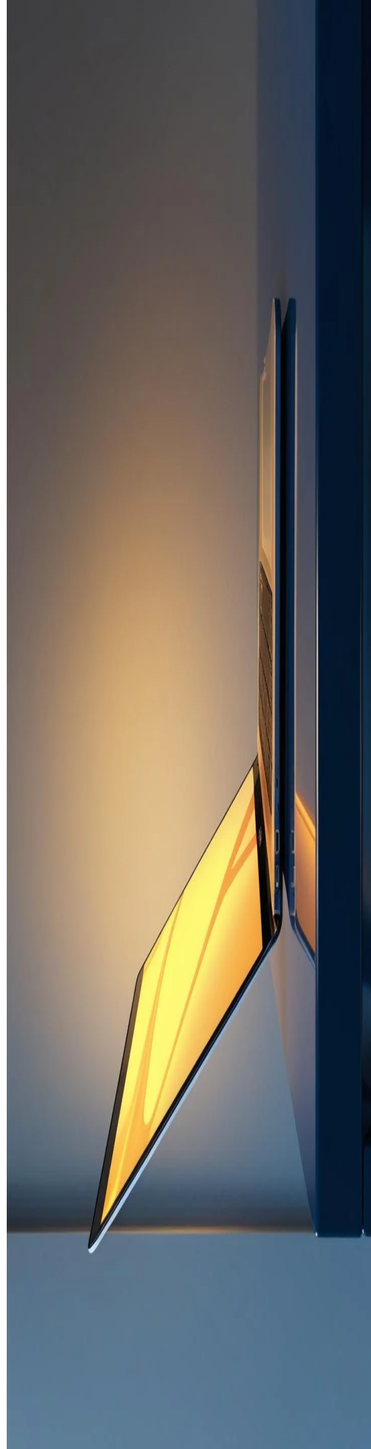
- **Pub/Sub model**: Publishers send messages to a topic, and the topic distributes them to its subscribers.
- **Multiple protocols**: Supports SMS, email, mobile push notifications, HTTP/S endpoints, AWS Lambda, Amazon SQS (fan-out), and Kinesis Data Firehose.
- **AWS service integration**: Many AWS services, such as Amazon CloudWatch and Amazon S3, can publish events or trigger notifications through SNS.





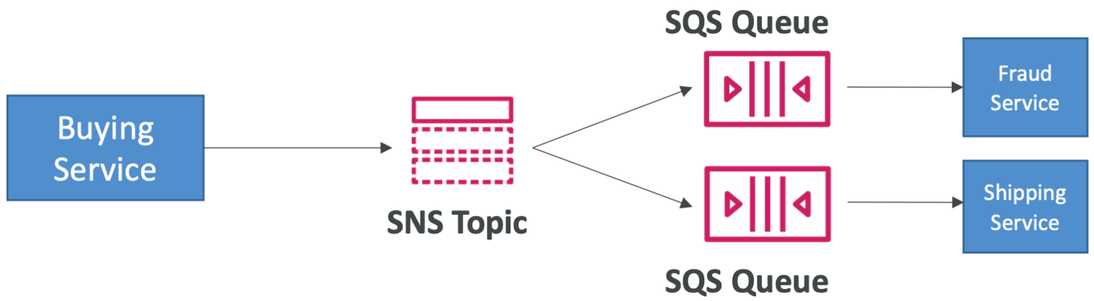
Amazon SNS features (continued)

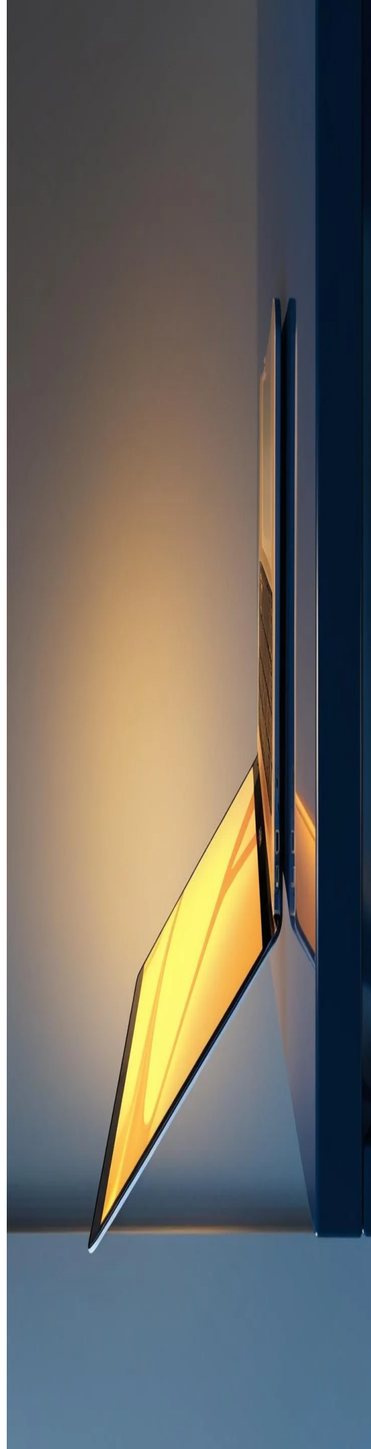
- **High throughput**: Sends millions of messages per second.
- **Message filtering**: Subscribers can receive only specific messages based on attributes.
- **Durability and retry logic**: SNS retries failed deliveries according to the delivery policy for supported endpoints. Delivery guarantees vary by protocol.
- **Encryption**: Supports AWS Key Management Service (KMS) for encryption at rest.
- **Simple APIs**: Makes it easy for developers to integrate messaging into applications.
- **Fully managed service**: AWS handles the underlying infrastructure, minimizing maintenance and management.
- **Pay-as-you-go pricing**: Costs are based on usage rather than requiring upfront infrastructure investment.
- **Scalability and flexibility**: Supports applications ranging from simple notifications to large distributed event-driven systems.



Amazon SNS: Uses Cases

- **Order notifications:** Send order confirmations and shipping updates to customers.
- **Alerts and notifications:** Send alerts through email or SMS.
- **Serverless applications:** Trigger AWS Lambda functions in response to application events.
- **Application integration:** Distribute notifications and events to multiple services.
- **Fan-out messaging (SNS+SQS):** Broadcast events to multiple SQS queues so different applications can process the same event independently. SNS provides asynchronous message distribution, while SQS buffers messages until each consumer is ready to process them. **Example:** An e-commerce order event is sent to separate SQS queues for inventory, shipping, and analytics.

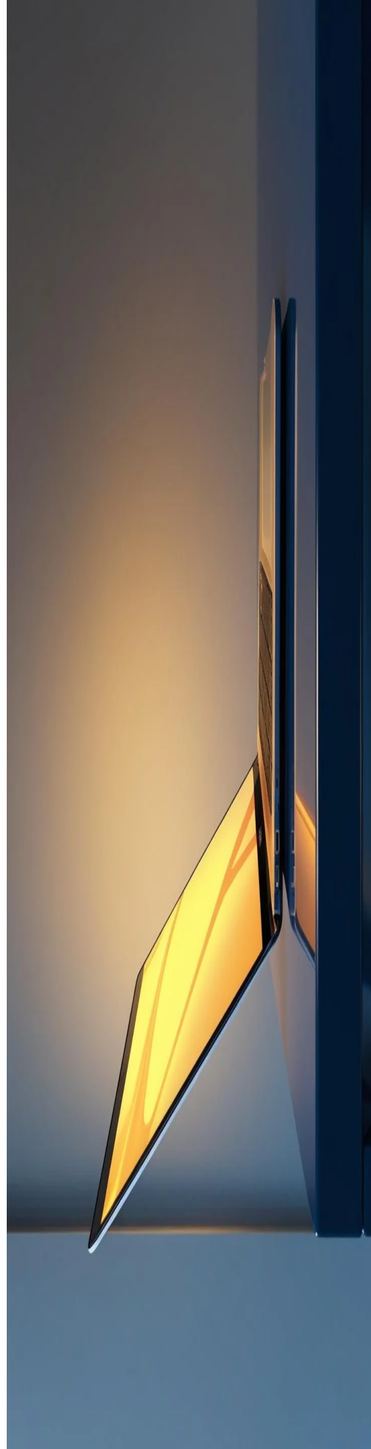




What is Amazon SES?

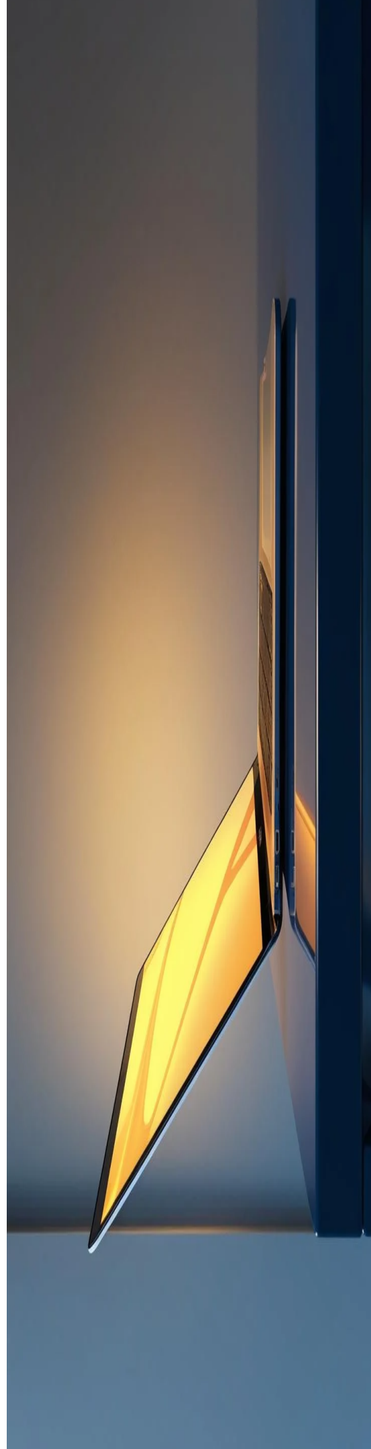
- Amazon SES is a fully managed email service that allows applications and businesses to send and receive emails securely, reliably, and at scale without managing their own mail servers.
- Main idea: Send and receive customized emails at scale.





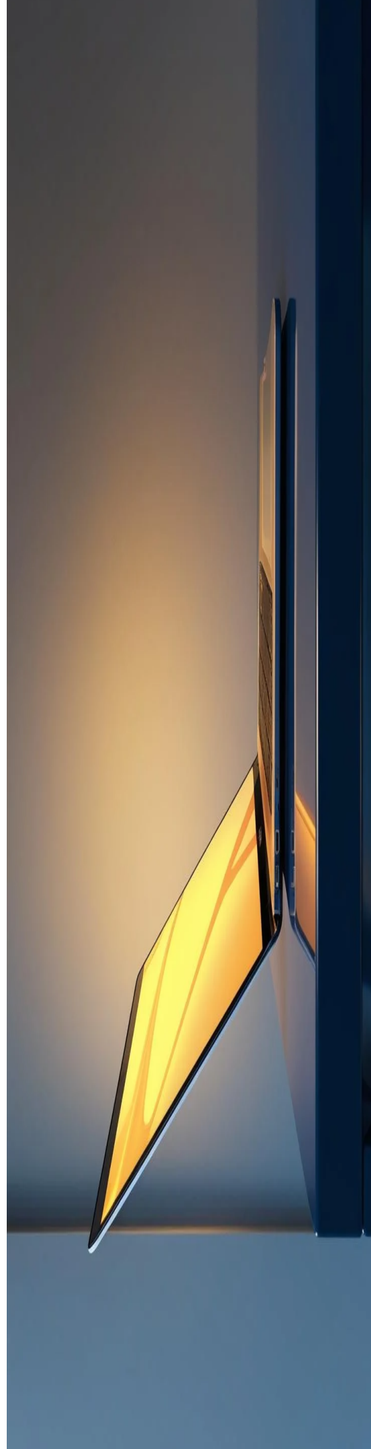
Amazon SES features

- **Send emails**: Send transactional emails (e.g., order confirmations), marketing campaigns, and bulk emails.
- **Receive emails**: Process incoming emails using integrations with Amazon S3, AWS Lambda, or Amazon SNS.
- **Deliverability insights**: Monitor email bounces, complaints, deliveries, and engagement metrics such as opens/clicks rates.
- **Security and compliance**: Supports domain authentication through SPF, DKIM, and DMARC, along with TLS encryption for email transmission.
- **Custom email templates**: Personalize emails using templates and variables, such as customer names and order numbers.
- **Pay-as-you-go pricing**: Pay based on usage, without needing to operate your own email servers.



Amazon SES: Uses Cases

- **Transactional emails:** Send order confirmations, receipts, password resets, and shipping updates.
- **Marketing emails:** Send newsletters, promotional campaigns, and product announcements.
- **Application notifications:** Send automated alerts, account verification emails, and system notifications.
- **Email integration:** Integrate email capabilities into websites, applications, and enterprise systems using APIs or SMTP.



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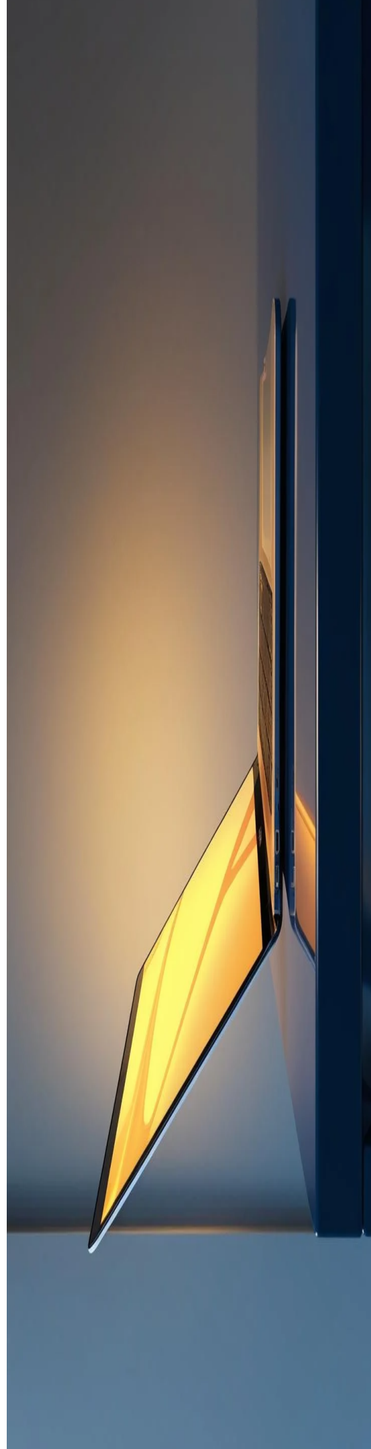
SNS vs. SES

Amazon SES

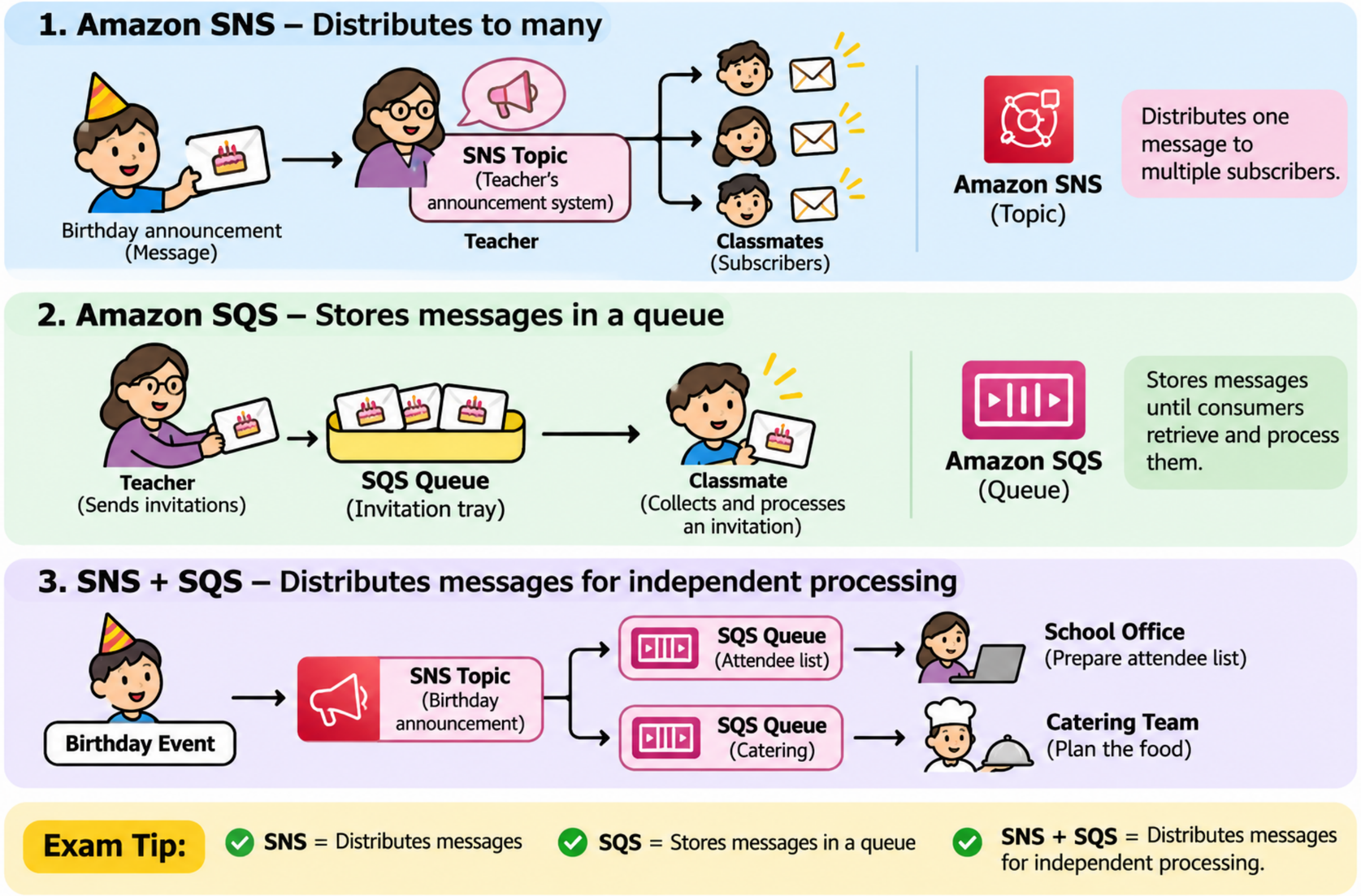
- Sends a beautifully designed HTML newsletter with images, buttons, branding, and personalized customer names.
- Best for email campaigns and customized emails.

Amazon SNS

- Sends an alert when a server's CPU usage exceeds 90%.
- Best for event-driven notifications across multiple channels.

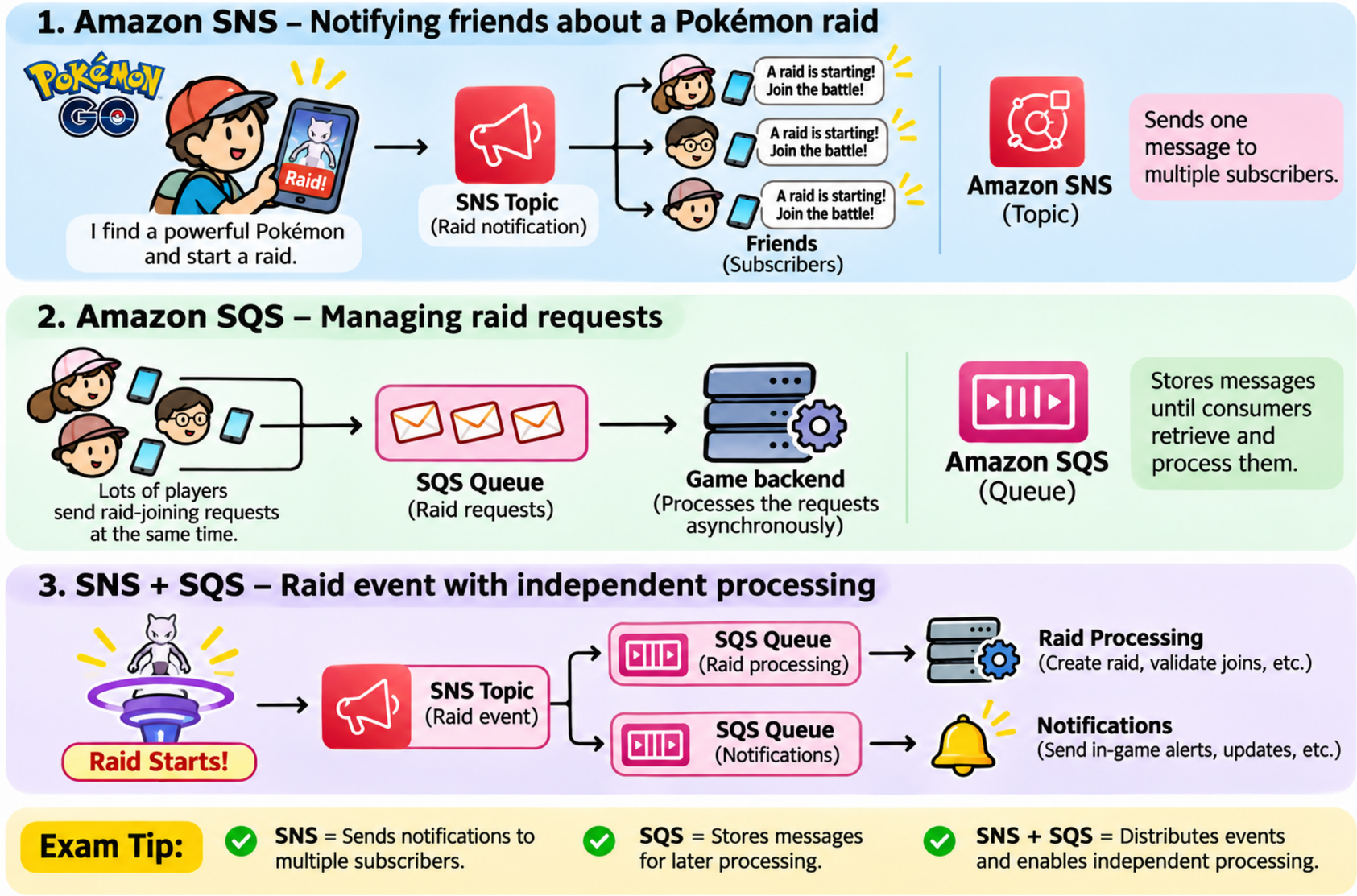


AWS SNS vs SQS — Birthday Invitation Example



https://github.com/chiaratardioli/aws-restart-journey/blob/master/Teaching/Topic/SQS_SNS_SES/SNS_SQS_Birthday_Invitation_Example.md

AWS SNS vs SQS — Pokémon GO Example



https://github.com/chiaratardioli/aws-restart-journey/blob/master/Teaching/Topic/SQS_SNS_SES/SNS_SQS_Pokemon_GO_Example.md

Summary

Amazon SQS (Simple Queue Service)

Purpose: Message queuing between applications.

- Stores messages in a queue until consumers process them.
- Enables asynchronous communication between services.
- Helps decouple applications.
- Types: Standard and FIFO queues.

Example: A gaming app queues player scores for background processing.



SQS

Amazon SNS (Simple Notification Service)

Purpose: Publish and distribute messages to multiple subscribers.

- Uses a publish/subscribe (pub/sub) model.
- Pushes notifications to subscribers.
- Supports endpoints such as SQS, AWS Lambda, HTTP/S, email, and SMS.

Example: Notify multiple services when a new order is placed.



SNS

Amazon SES (Simple Email Service)

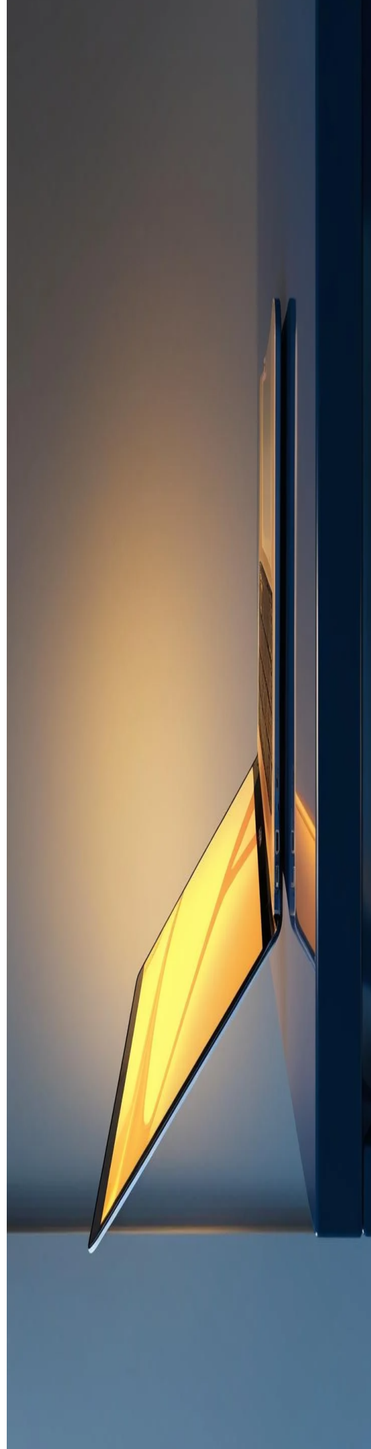
Purpose: Send and receive email.

- Sends transactional and bulk emails.
- Supports email verification and reputation management.
- Useful for application-generated emails.

Example: Send password-reset emails, receipts, and account confirmations.



SES

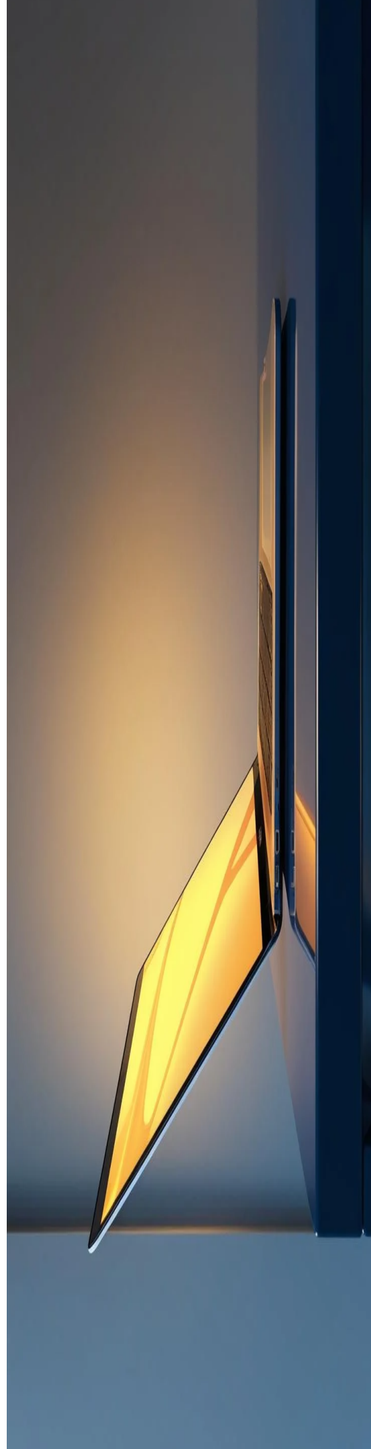


Questions

1. An application must send one event to three independent processing services, ensuring each service can process the event at its own pace. Which combination is most suitable?
    a. SES only
    b. SNS with multiple SQS queues
    c. One SQS queue shared by all services
    d. CloudWatch only

2. An application must store messages until a worker is ready to process them. Which service should you use?
    a. Amazon SNS
    b. Amazon SQS
    c. Amazon SES
    d. Amazon Route 53

3. A company wants to send a notification to multiple subscribers whenever a new order is placed. Which service is best?
    a. Amazon SNS
    b. Amazon EBS
    c. Amazon SES only
    d. AWS Config



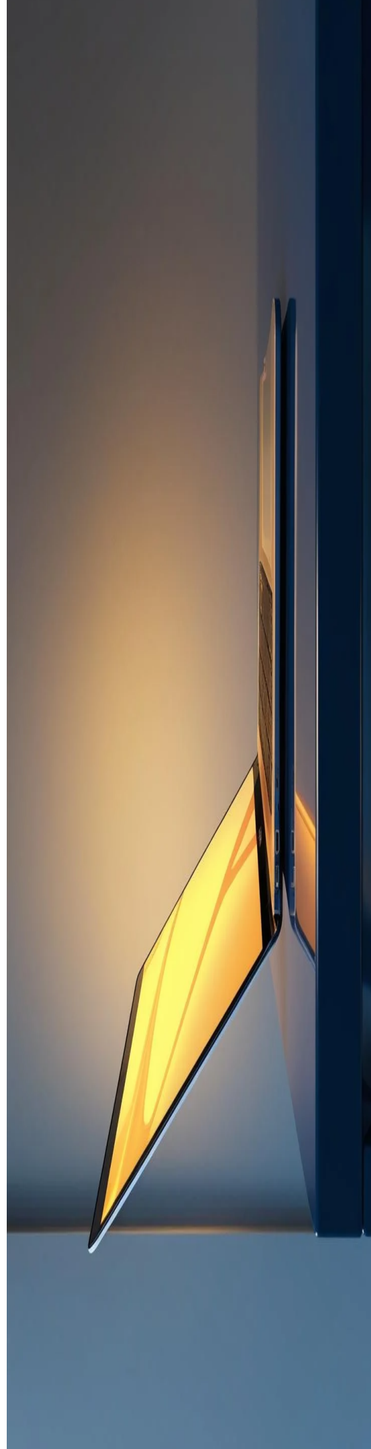
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Questions

4. Which SQS queue type supports ordering and exactly-once processing features?
    a. A. Standard
    b. FIFO
    c. SNS queue
    d. SES queue

5. An application must send an event to several independent services, and each service needs its own copy to process later. What is the best design?
    a. One shared SQS queue only
    b. SNS topic with multiple SQS queues
    c. SES with multiple email addresses
    d. One EC2 instance

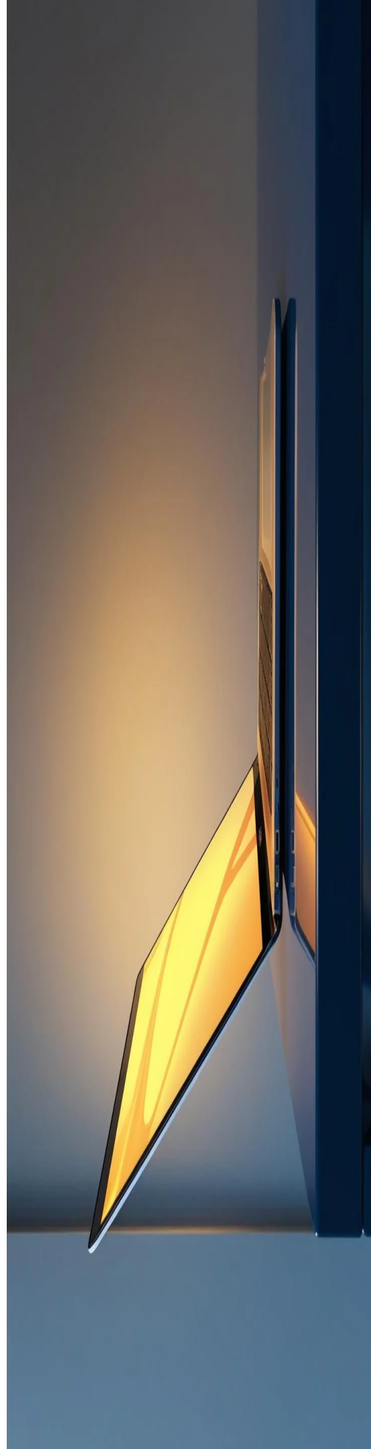
6. Which service is best for sending SMS notifications as well as application notifications to supported subscribers?
    a. Amazon SNS
    b. Amazon EFS
    c. Amazon Inspector
    d. Amazon S3



Questions

7. Two applications need asynchronous communication so neither has to wait for the other to finish. Which service is most appropriate?
    a. Amazon SQS
    b. Amazon SES
    c. AWS Shield
    d. Amazon CloudFront

8. A company needs an email delivery service for marketing campaigns and transactional emails. Which AWS service fits?
    a. Amazon SQS
    b. Amazon SES
    c. Amazon SNS only
    d. AWS CloudTrail



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THANK YOU.